

Medical Store Management System

Project Manual — Complete Reference

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BS Artificial Intelligence — 4th Semester

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12	5	4	4	7	30+
Database Tables	User Roles	Triggers	Stored Procs	SQL Views	Use Cases

React.js	Node.js / Express	MySQL 8.0	TailwindC SS	JWT Auth	jsPDF	Vite	ReportLab
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SEC-01 Project Overview

The Medical Store Management System (MSMS) is a full-stack web application built to digitise and streamline all operations of a pharmacy or medical store. It covers inventory management, sales processing, prescription tracking, supplier coordination, employee management, payment handling, and business analytics. The system is built using React.js on the frontend, Node.js/Express on the backend, and MySQL 8.0 as the primary relational database — normalized to Third Normal Form (3NF).

Objective ID	Description
OBJ-01	Design a 3NF-normalized relational database with 12 interrelated tables
OBJ-02	Implement all core MySQL features: DDL, DML, JOINS, subqueries, views, triggers, stored procedures, transactions
OBJ-03	Build a React.js frontend with role-based access control (5 roles)
OBJ-04	Implement a POS-style order interface with live cart calculation
OBJ-05	Generate printable bills and downloadable A5 PDF invoices
OBJ-06	Enforce full validation at frontend, backend, and database levels
OBJ-07	Provide management reports: low stock, expiry, revenue, top medicines
OBJ-08	Package as a GitHub-ready project with setup documentation

SEC-02 System Architecture

Layer ID	Layer	Technology	Responsibility
LAYER-01	Presentation	React.js + Vite + TailwindCSS	Browser UI, forms, bill preview, PDF download
LAYER-02	State	React Context + react-hook-form	Auth context, cart context, form validation
LAYER-03	API Client	Axios	HTTP calls to /api/v1/* endpoints
LAYER-04	Backend	Node.js + Express.js	REST API, JWT auth, business logic routing
LAYER-05	Validation	express-validator	Server-side input sanitisation and validation
LAYER-06	Auth	bcryptjs + jsonwebtoken	Password hashing, JWT issuance and verification
LAYER-07	Database	MySQL 8.0 via mysql2/promise	12 tables, triggers, stored procs, views, indexes
LAYER-08	PDF Engine	jsPDF + html2canvas (frontend)	Captures bill DOM node, exports as A5 PDF

SEC-03 Database Schema — 12 Tables

Table ID	Table Name	Primary Key	Purpose
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TBL-01	categories	category_id (PK)	Classifies medicines into groups (Antibiotic, Painkiller, etc.)
TBL-02	manufacturers	manufacturer_id (PK)	Pharmaceutical companies producing medicines
TBL-03	suppliers	supplier_id (PK)	Suppliers with ratings, contact info, and active flag
TBL-04	medicines	medicine_id (PK)	Master medicine catalog with price, dosage form, Rx flag
TBL-05	stock	stock_id (PK)	Quantity, batch, expiry per medicine — one-to-one with medicines
TBL-06	doctors	doctor_id (PK)	Licensed doctors for prescription validation
TBL-07	customers	customer_id (PK)	Customer profiles linked to orders and prescriptions
TBL-08	employees	employee_id (PK)	Staff records with role, salary, login credentials
TBL-09	prescriptions	prescription_id (PK)	Doctor-to-customer prescriptions with validity dates
TBL-10	orders	order_id (PK)	Master order with status, discount, tax, and total
TBL-11	order_items	item_id (PK)	Line items per order; subtotal GENERATED column
TBL-12	payments	payment_id (PK)	Payment method, status, transaction reference per order

All tables use surrogate integer primary keys (AUTO_INCREMENT). Foreign key constraints enforce referential integrity with appropriate ON DELETE CASCADE or RESTRICT behaviour. All numeric status and type fields use ENUM. Prices use DECIMAL(10,2). The database is normalized to 3NF — no transitive dependencies exist.

SEC-04 User Roles and Permissions

Role ID	Role	Access Level	Can Sell	Edit Inventory	Manage Staff	View Reports	System Config
ROLE-01	Admin	Full	Yes	Yes	Yes	Yes	Yes
ROLE-02	Manager	High	Yes	Yes	No	Yes	No
ROLE-03	Pharmacist	Medium	No	Yes	No	Partial	No
ROLE-04	Cashier	Limited	Yes	No	No	No	No
ROLE-05	Customer	None	—	—	—	—	—

SEC-05 Use Cases — Admin (ROLE-01)

Use Case ID	Actor	Title
UC-ADM-01	Admin	Login to system
Description		
Admin authenticates using username and password to access the full system.		
Precondition	Steps	Postcondition
Admin account exists in employees table	• Navigate to /login • Enter username and password • System validates credentials via bcryptjs • JWT token issued and stored in localStorage • Redirect to dashboard	Admin is authenticated; JWT stored

Use Case ID	Actor	Title
UC-ADM-02	Admin	Create employee account
Description		
Admin creates a new employee record with role, contact info, and login credentials.		
Precondition	Steps	Postcondition
Admin is logged in	• Go to Employees module • Click Add Employee • Fill name, role, phone, salary, username • System hashes password via bcryptjs • Record saved to employees table	New employee can log in with their credentials

Use Case ID	Actor	Title
UC-ADM-03	Admin	Assign or change employee role
Description		
Admin modifies a staff member's role to control their system access level.		
Precondition	Steps	Postcondition
Employee record exists; Admin logged in	• Open Employees list • Select employee • Click Edit • Change role field (ENUM dropdown) • Save; JWT role updated on next login	Employee's access level changes on their next session

Use Case ID	Actor	Title
UC-ADM-04	Admin	Delete employee account
Description		
Admin soft-deletes or permanently removes an employee account.		
Precondition	Steps	Postcondition
Admin is logged in; employee has no pending orders	• Open Employees list • Select employee • Click Delete • Confirm dialog appears • Set is_active = 0 (soft delete) or hard delete	Employee can no longer log in

Use Case ID	Actor	Title
UC-ADM-05	Admin	View and export all reports
Description		
Admin accesses all analytics views and exports data as CSV.		
Precondition	Steps	Postcondition
Admin is logged in	• Navigate to Reports • Select tab: Low Stock / Expiry / Sales / Top Medicines • View table data from MySQL views • Click Export CSV for any report	CSV file downloaded to browser

Use Case ID	Actor	Title
UC-ADM-06	Admin	Configure system settings
Description		
Admin sets store name, address, tax rate, DRAP license number used on bills.		
Precondition	Steps	Postcondition
Admin is logged in	• Go to Settings page • Edit store name, address, tax %, DRAP licence • Save; settings stored in DB config table • Bill preview reflects new values immediately	All future bills use updated store information

SEC-06 Use Cases — Manager (ROLE-02)

Use Case ID	Actor	Title
UC-MGR-01	Manager	Add new medicine to catalog
Description		
Manager adds a new medicine with dosage, price, and prescription flag.		
Precondition	Steps	Postcondition
Manager is logged in; category and manufacturer exist	• Go to Inventory > Medicines • Click Add Medicine • Fill name, generic name, category, manufacturer, dosage form, strength, price • Toggle requires_prescription if Rx-only • Submit; record saved to medicines table	Medicine appears in catalog and can be added to stock

Use Case ID	Actor	Title
UC-MGR-02	Manager	Manage supplier records
Description		
Manager adds, edits, or deactivates supplier records including ratings.		
Precondition	Steps	Postcondition
Manager is logged in	• Go to Suppliers module • Add or select supplier • Fill name, contact person, phone, email, rating (1.0–5.0) • Save; CHECK constraint enforces rating range	Supplier available for stock entries

Use Case ID	Actor	Title
UC-MGR-03	Manager	Approve restock order
Description		
Manager raises a restock request which calls the RestockMedicine stored procedure.		
Precondition	Steps	Postcondition
Manager logged in; medicine and supplier exist	• Go to Inventory > Stock • Click Restock on a medicine • Enter quantity, supplier, batch number, new expiry date • Submit; backend calls RestockMedicine() stored procedure • Stock quantity updated; stock_transactions log entry created	Stock quantity increased; log entry recorded

Use Case ID	Actor	Title
UC-MGR-04	Manager	View customer purchase history
Description		
Manager looks up a customer's full order and prescription history by phone number.		
Precondition	Steps	Postcondition
Manager logged in; customer exists	• Go to Customers • Search by phone number • Open customer profile • View orders tab and prescriptions tab	Full purchase and prescription history displayed

Use Case ID	Actor	Title
UC-MGR-05	Manager	Set reorder level per medicine
Description		
Manager configures the minimum stock threshold that triggers a low-stock alert.		
Precondition	Steps	Postcondition
Manager logged in; medicine and stock record exist	• Go to Inventory > Stock • Click edit on a medicine's stock row • Update reorder_level field • Save	Trigger will flag alert when quantity drops below new level

SEC-07 Use Cases — Pharmacist (ROLE-03)

Use Case ID	Actor	Title
UC-PH-01	Pharmacist	Verify and register prescription
Description		
Pharmacist logs a doctor's prescription for a customer before it can be used in a sale.		
Precondition	Steps	Postcondition
Pharmacist logged in; customer and doctor records exist	• Go to Prescriptions • Click Add Prescription • Select customer (search by phone) • Select doctor from directory • Enter prescription date and valid_until date • Add any notes; submit — saved to prescriptions table	Prescription available for Cashier to link to an order

Use Case ID	Actor	Title
UC-PH-02	Pharmacist	Check Rx requirement before dispensing
Description		
Pharmacist confirms whether a medicine requires a prescription before it can be sold.		
Precondition	Steps	Postcondition
Pharmacist logged in; medicine in catalog	• Search medicine in Inventory • Check requires_prescription flag • If flag is 1, notify Cashier that a valid prescription must be linked • System enforces this at order creation automatically	Rx-only medicines blocked from sale without linked prescription

Use Case ID	Actor	Title
UC-PH-03	Pharmacist	View expiry and stock levels
Description		
Pharmacist monitors medicines nearing expiry or below reorder threshold.		
Precondition	Steps	Postcondition
Pharmacist logged in	- Go to Reports - Select Low Stock tab — queries vw_LowStockAlert - Select Expiring Soon tab — queries vw_ExpiringSoon - Review list and notify Manager for action	Pharmacist aware of critical stock situations

Use Case ID	Actor	Title
UC-PH-04	Pharmacist	Add new doctor to directory
Description		
Pharmacist registers a licensed doctor whose prescriptions the store accepts.		
Precondition	Steps	Postcondition
Pharmacist logged in	- Go to Doctors module - Click Add Doctor - Fill name, specialization, license number (UNIQUE), hospital, phone - Save; UNIQUE constraint on license_number prevents duplicates	Doctor available in prescription registration dropdown

SEC-08 Use Cases — Cashier (ROLE-04)

Use Case ID	Actor	Title
UC-CSH-01	Cashier	Create a new sale order
Description		
Cashier opens the POS interface and builds a cart for a customer transaction.		
Precondition	Steps	Postcondition
Cashier logged in; at least one medicine in stock	• Go to Sales > New Order • Search medicine by name in search box • Click Add to Cart; set quantity • System checks stock availability in real time • Repeat for all items in the sale	Cart populated with items; subtotal calculated live

Use Case ID	Actor	Title
UC-CSH-02	Cashier	Link customer to order
Description		
Cashier attaches a registered customer or creates a new profile for the transaction.		
Precondition	Steps	Postcondition
POS interface open	• Search customer by phone number • If found — select from dropdown • If not found — toggle 'Register new customer', fill name + phone • Or toggle 'Walk-in' to skip customer linkage	Order linked to customer_id or marked as walk-in (customer_id NULL)

Use Case ID	Actor	Title
UC-CSH-03	Cashier	Handle prescription-only medicines
Description		
When an Rx medicine is in the cart, Cashier must link a valid prescription.		
Precondition	Steps	Postcondition
Cart contains a medicine with requires_prescription = 1	• System shows warning banner: 'Prescription required' • Cashier searches prescriptions by customer phone • Selects valid, unused prescription from dropdown • System checks valid_until date >= today • Order cannot be placed until prescription is linked	Prescription linked; is_used set to 1 after order completes

Use Case ID	Actor	Title
UC-CSH-04	Cashier	Apply discount and process payment
Description		
Cashier applies optional discount, selects payment method, and finalises the sale.		
Precondition	Steps	Postcondition
Cart has items; customer linked or walk-in selected	• Enter discount % (0–100); grand total recalculates live • Select payment method: Cash / Card / Online / Insurance • Click 'Place Order & Pay' • Backend runs PlaceOrder() then CompletePayment() in a transaction • Triggers auto-deduct stock and recalculate order total • Redirect to Bill Preview page	Order status = Completed; payment recorded; stock updated

Use Case ID	Actor	Title
UC-CSH-05	Cashier	Print bill
Description		
Cashier prints the on-screen bill for the customer using the browser print dialog.		
Precondition	Steps	Postcondition
Order completed; Bill Preview page loaded	- Click Print Bill button on Bill Preview page - window.print() triggers browser print dialog - CSS @media print hides sidebar, navbar, and all buttons - Only BillPreview component is visible in print output - Customer receives physical receipt	Physical bill printed with full order details

Use Case ID	Actor	Title
UC-CSH-06	Cashier	Download PDF invoice
Description		
Cashier or customer downloads the bill as a professionally formatted A5 PDF.		
Precondition	Steps	Postcondition
Order completed; Bill Preview page loaded	- Click Download PDF button - html2canvas captures the BillPreview div as a canvas image - jsPDF creates an A5-sized PDF document - Canvas image embedded into PDF - Browser downloads file: Invoice-ORD-00001-2026-03-12.pdf	PDF saved to local device; can be emailed or reprinted

SEC-09 Use Cases — Customer (ROLE-05 — external)

Use Case ID	Actor	Title
UC-CUS-01	Customer	Purchase medicines over the counter
Description		
Customer visits the store, requests medicines, and completes a cash or card transaction.		
Precondition	Steps	Postcondition
Customer is physically present at the store	• Customer states medicines needed to Cashier • Cashier builds the cart (UC-CSH-01) • Customer provides phone number for profile lookup or registration • Customer pays; receives printed bill	Transaction recorded; stock updated; bill issued

Use Case ID	Actor	Title
UC-CUS-02	Customer	Submit prescription for Rx medicines
Description		
Customer provides a valid doctor's prescription to purchase Rx-only medicines.		
Precondition	Steps	Postcondition
Customer has a physical prescription from a registered doctor	• Customer hands prescription to Pharmacist • Pharmacist logs it in the system (UC-PH-01) • Cashier links prescription to the order (UC-CSH-03) • Medicine dispensed after validation	Prescription recorded and marked used; medicine dispensed

Use Case ID	Actor	Title
UC-CUS-03	Customer	Receive and download PDF receipt
Description		
Customer receives a digital PDF invoice via download or email.		
Precondition	Steps	Postcondition
Order is completed; Cashier on Bill Preview page	- Cashier clicks Download PDF (UC-CSH-06) - PDF generated (A5, jsPDF) - Customer can scan a QR or receive PDF on their phone	Customer has digital proof of purchase

SEC-10 Use Cases — System / Database (automated)

Use Case ID	Actor	Title
UC-SYS-01	System	Auto-deduct stock on sale
Description		
MySQL trigger automatically reduces stock quantity when an order item is inserted.		
Precondition	Steps	Postcondition
Order item inserted into order_items table	- AFTER INSERT trigger fires on order_items - UPDATE stock SET quantity = quantity - NEW.quantity WHERE medicine_id = NEW.medicine_id - If quantity would go below 0, SIGNAL SQLSTATE '45000' — order blocked	Stock reduced; prevents negative stock automatically

Use Case ID	Actor	Title
UC-SYS-02	System	Flag low stock alert
Description		
After any stock deduction, system checks if quantity dropped below reorder level.		
Precondition	Steps	Postcondition
UC-SYS-01 trigger completed	• Second AFTER INSERT trigger checks stock.quantity < stock.reorder_level • If true, INSERT into low_stock_alerts table • Dashboard reads this table and shows alert banner	Alert visible on dashboard; Manager notified

Use Case ID	Actor	Title
UC-SYS-03	System	Recalculate order total
Description		
Trigger recalculates subtotal, applies discount and tax, updates orders.total_amount.		
Precondition	Steps	Postcondition
Order item inserted or updated	• AFTER INSERT trigger on order_items fires • SELECT SUM(subtotal) from order_items WHERE order_id = NEW.order_id • Apply discount_percent and tax_percent from orders table • UPDATE orders SET subtotal = ..., total_amount = ... WHERE order_id = NEW.order_id	Order total always accurate without manual calculation

SEC-11 Bill & PDF Feature

The bill system is the most customer-facing feature of MSMS. Every completed order produces a formatted bill that can be printed or saved as a PDF.

Field ID	Field	Details
BILL-01	Store header	Name, address, phone, DRAP licence number
BILL-02	Invoice number	ORD-XXXXXX format, auto-incremented order ID
BILL-03	Date and time	Order timestamp formatted DD-Mon-YYYY HH:MM
BILL-04	Cashier name	Employee name from JWT session
BILL-05	Customer info	Name + phone; 'Walk-in Customer' if unregistered
BILL-06	Prescription	Rx number shown if prescription was linked
BILL-07	Line items	#, Medicine Name, Qty, Unit Price, Subtotal per item
BILL-08	Subtotal	Sum of all item subtotals in PKR
BILL-09	Discount	Discount % and PKR amount deducted
BILL-10	Tax	Tax % and PKR amount added
BILL-11	Grand Total	Final amount in PKR, bold and large
BILL-12	Payment method	Cash / Card / Online / Insurance + PAID stamp
BILL-13	Footer	Thank you message + return policy (3 days with receipt)
BILL-14	Print action	window.print() with @media print CSS — hides all UI chrome
BILL-15	PDF action	html2canvas + jsPDF → A5 PDF → filename: Invoice-ORD-XXXXXX-YYYY-MM-DD.pdf

SEC-12 Triggers Reference

Trigger ID	Name	Timing	Table	Purpose
TRG-01	after_order_item_insert	AFTER INSERT	order_items	Deducts NEW.quantity from stock; signals error if stock would go negative
TRG-02	after_order_item_lowstock	AFTER INSERT	order_items	Inserts a row into low_stock_alerts if quantity < reorder_level post-deduction
TRG-03	after_order_item_update_total	AFTER INSERT	order_items	Recalculates orders.subtotal and total_amount applying discount and tax
TRG-04	before_stock_update_expiry_check	BEFORE UPDATE	stock	Signals error if NEW.expiry_date < CURDATE() — prevents expired stock entry

SEC-13 Stored Procedures Reference

Proc ID	Signature	Parameters	Description
SP-01	GenerateBill(order_id)	IN: order_id INT	Returns full bill dataset — order, items, medicine names, customer, employee, payment in one query
SP-02	RestockMedicine(medicine_id, qty, supplier_id, expiry_date, batch)	IN: 5 parameters	Validates inputs, updates stock quantity, updates supplier/batch/expiry, logs to stock_transactions
SP-03	PlaceOrder(customer_id, employee_id, prescription_id, discount, tax, notes, OUT order_id)	IN: 6 OUT: 1	Opens transaction, inserts order row, returns new order_id for item insertion
SP-04	CompletePayment(order_id, method, amount, ref)	IN: 4 parameters	Wraps INSERT into payments + UPDATE orders status = Completed in one atomic transaction

SEC-14 Views Reference

View ID	View Name	Description
VW-01	vw_LowStockAlert	Medicines where stock.quantity < stock.reorder_level; includes category name
VW-02	vw_ExpiringSoon	Medicines expiring within 30 days from CURDATE(); sorted by expiry_date ASC
VW-03	vw_SalesSummary	Completed orders grouped by DATE(order_date) → sale_date, total_orders, total_revenue
VW-04	vw_TopSellingMedicines	SUM(order_items.quantity) grouped by medicine, ordered DESC
VW-05	vw_EmployeePerformance	COUNT(orders) and SUM(total_amount) per employee for performance tracking
VW-06	vw_SupplierRatings	Suppliers ranked by rating DESC with stock delivery count
VW-07	vw_FullInventory	Medicines + stock + category + manufacturer + supplier — complete inventory join

SEC-15 API Endpoints

API ID	Method	Endpoint	Access	Description
API-01	POST	/api/v1/auth/login	Public	Employee login → returns JWT
API-02	GET	/api/v1/medicines	All roles	List medicines (search, pagination)
API-03	POST	/api/v1/medicines	Mgr+	Add new medicine
API-04	PUT	/api/v1/medicines/:id	Mgr+	Update medicine
API-05	DELETE	/api/v1/medicines/:id	Mgr+	Soft-delete medicine
API-06	GET	/api/v1/stock	All roles	Get stock levels
API-07	PUT	/api/v1/stock/:id/restock	Mgr+/Ph	Call RestockMedicine()
API-08	GET	/api/v1/customers	Mgr+/CSH	List or search customers
API-09	POST	/api/v1/customers	Mgr+/CSH	Register new customer
API-10	GET	/api/v1/prescriptions	Mgr+/Ph/CSH	List prescriptions
API-11	POST	/api/v1/prescriptions	Ph+	Register prescription
API-12	POST	/api/v1/orders	CSH+	Call PlaceOrder()
API-13	POST	/api/v1/orders/:id/items	CSH+	Add item to order
API-14	POST	/api/v1/orders/:id/complete	CSH+	Call CompletePayment()
API-15	GET	/api/v1/orders/:id/bill	CSH+	Call GenerateBill()
API-16	GET	/api/v1/reports/low-stock	Mgr+	Query vw_LowStockAlert
API-17	GET	/api/v1/reports/expiring-soon	Mgr+/Ph	Query vw_ExpiringSoon
API-18	GET	/api/v1/reports/sales-summary	Mgr+	Query vw_SalesSummary
API-19	GET	/api/v1/reports/top-medicines	Mgr+	Query vw_TopSellingMedicines
API-20	GET	/api/v1/employees	Admin	List all employees
API-21	POST	/api/v1/employees	Admin	Create employee account
API-22	PUT	/api/v1/employees/:id	Admin	Update employee / change role
API-23	GET	/api/v1/suppliers	Mgr+	List suppliers
API-24	POST	/api/v1/suppliers	Mgr+	Add supplier
API-25	GET	/api/v1/doctors	Ph+	List doctors directory

SEC-16 Validation Rules

Rule ID	Field	Validation
VAL-01	Medicine name	Required; min 2 chars; max 200 chars
VAL-02	Unit price	Required; numeric; > 0; max 2 decimal places; DECIMAL(10,2)
VAL-03	Expiry date	Required; must be a future date (> CURDATE())
VAL-04	Phone number	Pakistani format: 11 digits starting with 03 or 92
VAL-05	Email	Standard RFC 5322 regex; UNIQUE in employees and customers
VAL-06	Discount percent	Numeric; 0–100; CHECK constraint in DB
VAL-07	Quantity (order)	Integer; > 0; CHECK constraint in order_items
VAL-08	Stock quantity	Integer; >= 0; cannot go negative (trigger blocks it)
VAL-09	Supplier rating	DECIMAL(3,1); CHECK between 1.0 and 5.0
VAL-10	Prescription date	valid_until >= prescription_date; CHECK constraint in DB
VAL-11	Rx-gated sale	If cart has Rx medicine, prescription_id must be linked; frontend and backend enforce
VAL-12	Password	Min 8 chars; hashed with bcryptjs (10 salt rounds) before storage
VAL-13	Salary	Numeric; > 0; CHECK constraint in employees table
VAL-14	Manufacturing date	Stock expiry_date > manufacturing_date; CHECK constraint in DB

SEC-17 Installation Guide

Step ID	Step	Command / Action
STEP-01	Prerequisites	Install Node.js 18+, MySQL 8.0, npm. Clone the repo from GitHub.
STEP-02	Root install	Run: npm install in the project root to install concurrently.
STEP-03	Backend setup	cd backend && npm install && cp .env.example .env — fill in DB credentials.
STEP-04	Frontend setup	cd ../frontend && npm install
STEP-05	Database: create	In MySQL: CREATE DATABASE msms_db; USE msms_db;
STEP-06	Database: schema	Run: source database/schema.sql
STEP-07	Database: data	Run: source database/sample_data.sql (10+ rows per table, PKR prices)
STEP-08	Database: logic	Run in order: triggers.sql → stored_procedures.sql → views.sql → indexes.sql
STEP-09	Start dev server	From project root: npm run dev (runs frontend on :5173, backend on :5000)
STEP-10	Default login	Username: admin Password: admin123 Role: Admin

The .env file must never be committed to GitHub. It is listed in .gitignore. The .env.example file (with no real credentials) is safe to commit.

SEC-18 File and Folder Structure

Dir ID	Path	Contents
DIR-01	msms/	Project root — package.json for concurrently
DIR-02	msms/backend/	Node.js + Express backend application
DIR-03	msms/backend/routes/	12 route files — one per resource (medicines, orders, etc.)
DIR-04	msms/backend/middleware/	auth.js (JWT) and validate.js (express-validator)
DIR-05	msms/backend/.env	DB credentials and JWT secret — NOT committed to Git
DIR-06	msms/frontend/	React.js + Vite application
DIR-07	msms/frontend/src/pages/	Page components: Dashboard, NewOrder, BillPreview, Reports, etc.
DIR-08	msms/frontend/src/components/	Shared UI: Sidebar, Modal, Table, StatCard, BillPreview, BillActions
DIR-09	msms/frontend/src/context/	AuthContext.jsx (JWT) and CartContext.jsx (POS cart state)
DIR-10	msms/frontend/src/api/	Axios instance with base URL and JWT interceptor
DIR-11	msms/database/	All SQL files: schema, sample_data, triggers, stored_procedures, views, indexes
DIR-12	msms/README.md	Full setup guide, API docs, default credentials, screenshots
DIR-13	msms/.gitignore	Excludes node_modules/, .env, dist/, *.log, .DS_Store

SEC-19 Glossary

Term ID	Term	Definition
GLO-01	3NF	Third Normal Form — database design with no transitive dependencies
GLO-02	JWT	JSON Web Token — stateless authentication token passed in HTTP headers
GLO-03	POS	Point of Sale — the interface where Cashier processes a customer transaction
GLO-04	Rx	Prescription — abbreviation for medicines that legally require a doctor's prescription
GLO-05	DRAP	Drug Regulatory Authority of Pakistan — the regulator for medicines in Pakistan
GLO-06	Trigger	A MySQL function that automatically executes in response to an INSERT/UPDATE/DELETE
GLO-07	Stored Procedure	A precompiled SQL routine stored in the database and called by name from the backend
GLO-08	View	A virtual table in MySQL defined by a SELECT query; used for reporting
GLO-09	Soft Delete	Setting is_active = 0 instead of removing the row — preserves history
GLO-10	DDL	Data Definition Language — SQL for creating tables, views, indexes (CREATE, ALTER, DROP)
GLO-11	DML	Data Manipulation Language — SQL for reading/writing data (SELECT, INSERT, UPDATE, DELETE)
GLO-12	bcryptjs	Node.js library that hashes passwords with a salt before storing them in the database
GLO-13	jsPDF	JavaScript library that creates PDF files in the browser from HTML content
GLO-14	html2canvas	JavaScript library that converts a DOM element to a canvas image for PDF embedding
GLO-15	Vite	A modern frontend build tool for React — faster than create-react-app
GLO-16	PKR	Pakistani Rupee — the currency used for all prices in the system
GLO-17	ENUM	A MySQL column type that restricts values to a defined list (e.g. 'Cash','Card','Online')
GLO-18	GENERATED column	A MySQL column whose value is automatically computed from other columns (subtotal = qty * price)

This manual covers all major components of the MSMS project. For the complete source code, SQL files, and setup instructions, refer to the GitHub repository README.md included in the project ZIP.